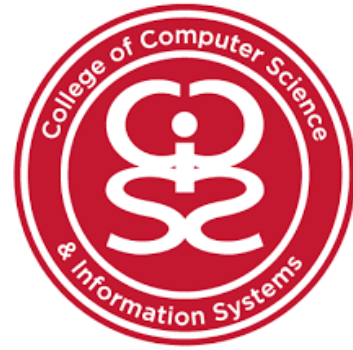




DISEASE OUTBREAK PREDICTION

(Time Series Forecasting of Global COVID-19 Cases with ARIMA)

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ABSTRACT

This project presents an end-to-end data science solution for forecasting global COVID-19 case trends using time-series analysis. Historical case data from the Johns Hopkins COVID-19 dataset was collected, cleaned, and transformed into a structured time-series format. Exploratory data analysis was performed to identify reporting patterns and long-term trends. Statistical techniques, including the Augmented Dickey-Fuller (ADF) test, were applied to evaluate stationarity, while Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots were used to determine appropriate model parameters.

An ARIMA(7,1,0) forecasting model was developed and trained to predict future outbreak trends. The trained model was serialized using Joblib and deployed through an interactive Streamlit dashboard, allowing users to generate outbreak forecasts for customizable time horizons. This project demonstrates the complete data science lifecycle, including data acquisition, preprocessing, statistical analysis, model development, visualization, and deployment.

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CHAPTER 1: INTRODUCTION

- **Background:**

The COVID-19 pandemic highlighted the importance of data-driven decision-making in healthcare. Disease forecasting helps authorities predict future case trends, allocate resources efficiently, and prepare for potential outbreaks. Time-series analysis provides an effective approach for identifying patterns in historical data and generating future predictions.

- **1.2 Importance of Disease Forecasting:**

Accurate outbreak forecasting can support hospital planning, resource allocation, and public health interventions. Predictive models enable organizations to take proactive measures rather than reacting after cases begin to rise.

- **1.3 Project overview:**

This project uses the Johns Hopkins COVID-19 dataset to forecast future case trends. Data preprocessing, statistical analysis, and ARIMA modeling were performed to generate predictions. The final model was deployed through a Streamlit dashboard that allows users to visualize future outbreak forecasts.

CHAPTER 2: PROBLEM STATEMENT

- **2.1 Problem Description:**

Disease outbreak prediction is challenging because case counts change over time and often contain trends, seasonal patterns, and reporting inconsistencies. These characteristics make accurate forecasting difficult using traditional methods.

- **2.2 Challenges in Disease Forecasting:**

Key challenges include non-stationary data, weekly reporting cycles, and sudden fluctuations in case counts. These issues can reduce forecasting accuracy if not properly addressed.

- **2.3 Proposed Solution:**

To address these challenges, an ARIMA (7,1,0) model was developed. Stationarity was verified using the Augmented Dickey-Fuller (ADF) test, while ACF and PACF plots were used to determine suitable model parameters. The trained model was then deployed using Streamlit.

CHAPTER 3: OBJECTIVES

• 3.1 Primary Objectives:

- The objectives of this project are:
- Collect and preprocess COVID-19 case data.
- Perform exploratory data analysis.
- Apply stationarity testing using the ADF test.
- Build an ARIMA forecasting model.
- Evaluate forecasting performance.
- Deploy the model through a Streamlit dashboard.

• 3.2 Learning Outcomes:

Through this project, we gained practical experience in data preprocessing, time-series analysis, ARIMA forecasting, data visualization, and Streamlit deployment. The project also strengthened our understanding of applying data science techniques to real-world healthcare problems.

CHAPTER 4: DATASET DESCRIPTION

- **4.1 Dataset Source:**

The dataset used in this project was the Johns Hopkins University COVID-19 Global Confirmed Cases dataset. It contains daily confirmed COVID-19 case counts reported from countries around the world and is widely used for outbreak analysis and forecasting research.

- **4.2 Dataset Structure:**

The dataset consists of daily observations of confirmed COVID-19 cases. Each row represents a specific date, while numerical columns contain case counts. For forecasting purposes, global daily case counts were aggregated into a single time-series variable called `Global_Cases`.

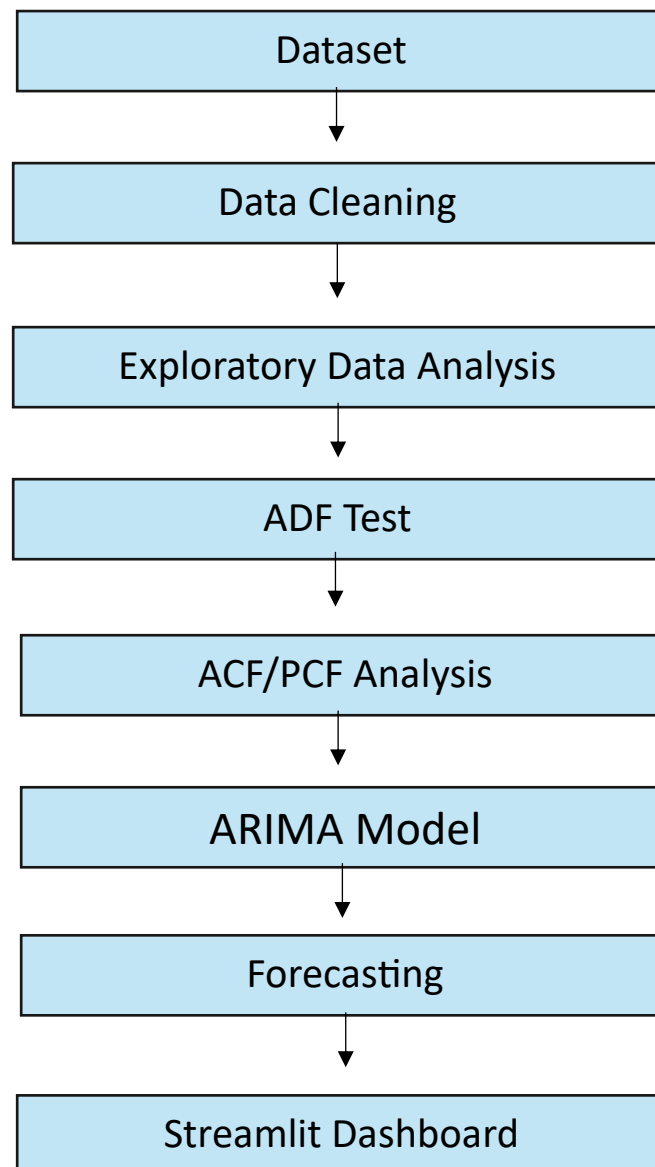
- **4.3 Data Preprocessing:**

Several preprocessing steps were performed before analysis. The date column was converted into datetime format, unnecessary rows were removed, and daily global case counts were calculated by summing confirmed cases across all countries. The processed data was then transformed into a time-series format suitable for forecasting.

CHAPTER 5: METHODOLOGY

• 5.1 Project workflow:

The project followed a structured data science workflow consisting of data collection, preprocessing, exploratory analysis, stationarity testing, model development, forecasting, and deployment.



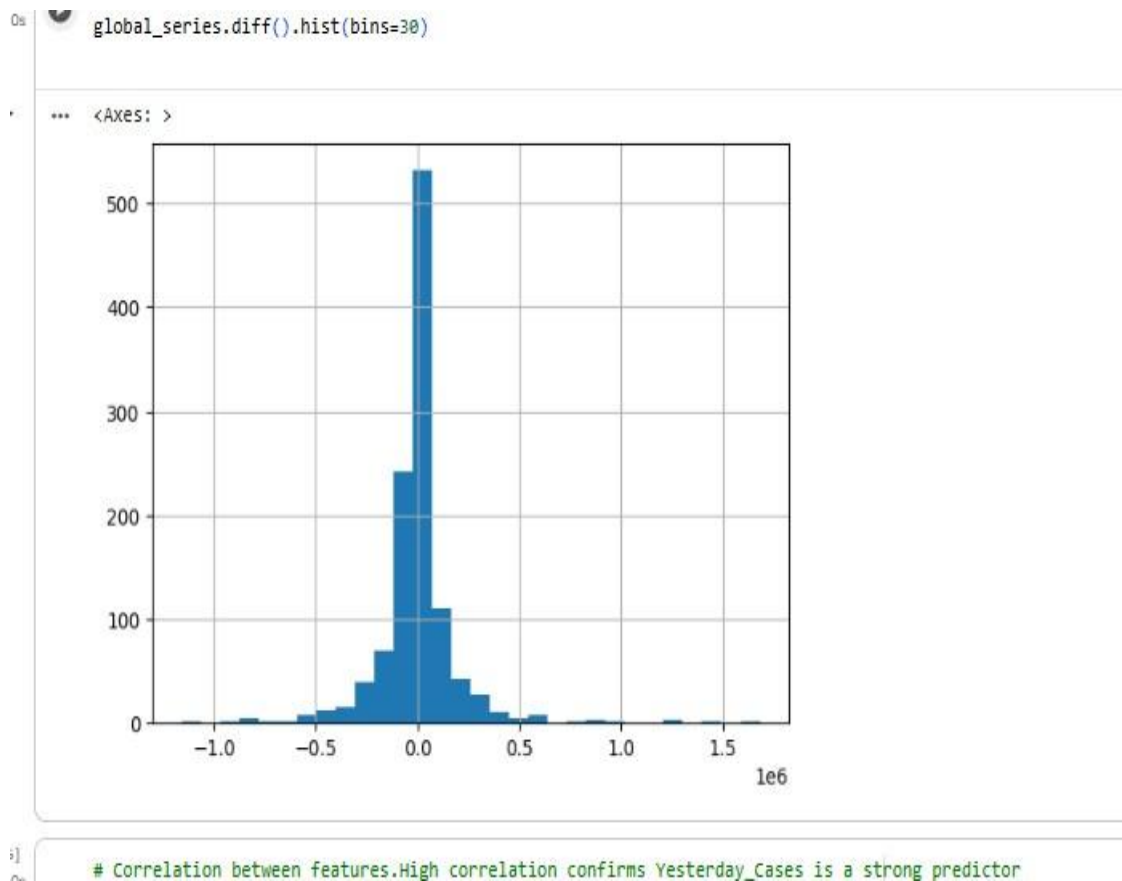
- **5.2 Tools and technologies:**

The project was developed using Python and several data science libraries. Pandas and NumPy were used for data manipulation, Matplotlib for visualization, Statsmodels for ARIMA modeling, Scikit-learn for Linear Regression, Joblib for model serialization, and Streamlit for deployment.

- **5.3 Exploratory Data Analysis (EDA):**

Exploratory Data Analysis was performed to understand the behavior of the dataset and identify trends. Visualizations showed significant fluctuations in case counts and revealed overall growth patterns throughout the pandemic per

- Histogram:



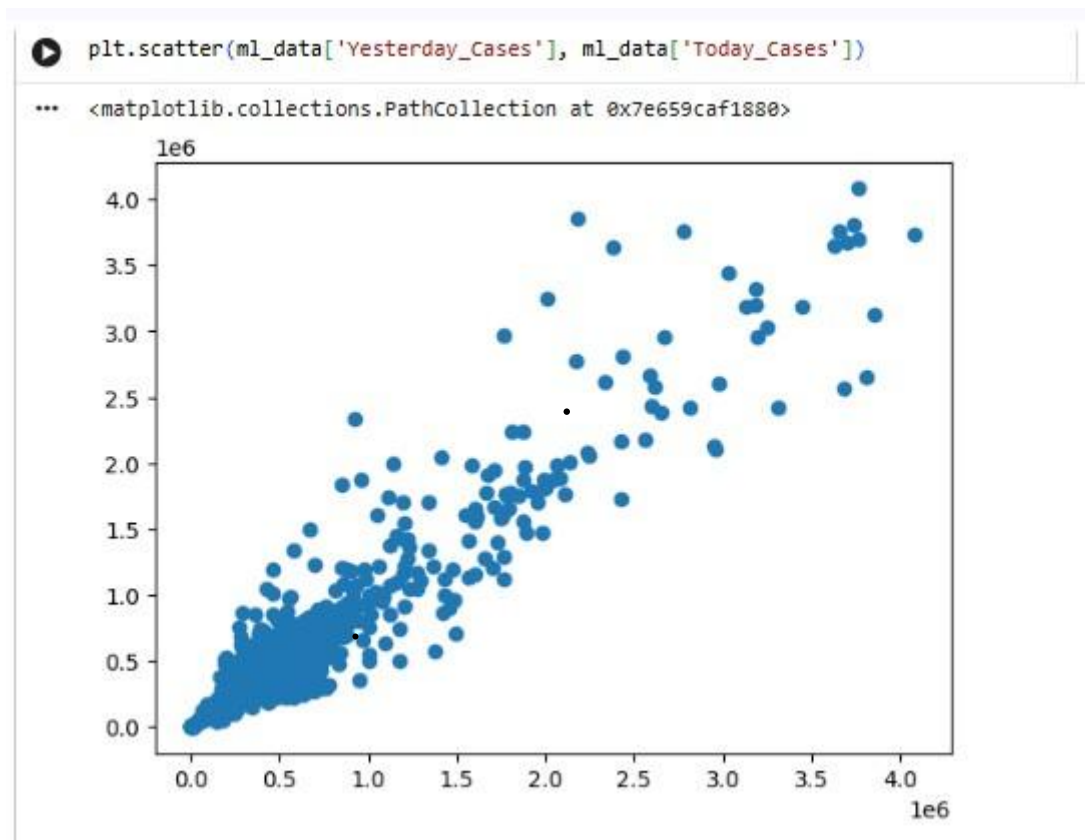
A histogram was generated to analyze the distribution of COVID-19 case counts. The graph shows variations in reported cases and highlights the presence of periods with significantly higher case numbers compared to normal observations.

- Heatmap:



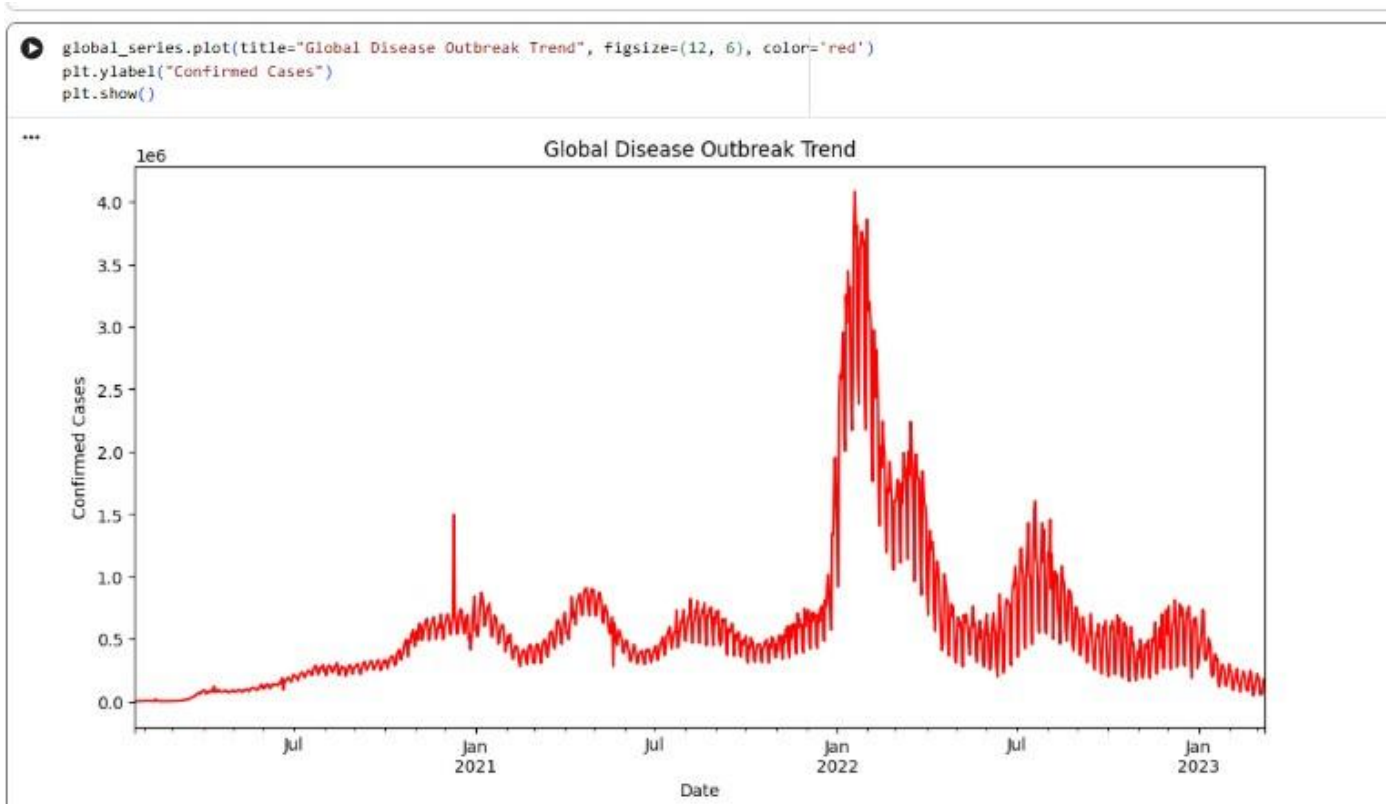
A correlation heatmap was used to visualize relationships between numerical variables in the dataset. Strong positive correlations indicate that variables tend to increase together, providing useful insights into underlying data patterns.

- Scatter plot:



The scatter plot illustrates the relationship between previous and current case counts. The upward trend of the data points suggests a positive relationship between consecutive observations, indicating that historical values can help predict future trends.

- Global Cases Trend Analysis:



The global cases trend graph shows how COVID-19 cases changed over time. A significant upward trend can be observed, along with fluctuations that reflect different phases of the pandemic. This visualization provides insight into the overall behavior of the outbreak.

CHAPTER 6: MODEL DEVELOPMENT:

- **6.1 ARIMA Model:**

The ARIMA(p, d, q) model was configured based on the stationarity and correlation analysis performed in the previous chapter. The original series was found to be borderline non-stationary, with a trending cumulative structure typical of outbreak data, so first-order differencing ($d = 1$) was applied to stabilize the trend. The autoregressive order ($p = 7$) and moving average order ($q = 0$) were selected manually from the PACF and ACF plots respectively, resulting in the final model ARIMA(7, 1, 0).

The ARIMA(7,1,0) model was selected based on the results of stationarity testing and the analysis of ACF and PACF plots. ARIMA is capable of capturing temporal dependencies and short-term patterns present in historical data through its autoregressive component. Due to its simplicity and effectiveness, ARIMA was considered suitable for demonstrating time-series forecasting techniques within the scope of this study.

- **6.2 Linear Regression Baseline:**

Linear Regression was implemented as a baseline forecasting model to provide a simple benchmark for comparison with the ARIMA model. In this approach, time was treated as the independent variable, while daily confirmed case counts served as the dependent variable. The dataset was divided into training and testing sets to evaluate predictive performance. Although Linear Regression is not specifically designed for time-series forecasting, it was included to assess how a simple trend-based model performs compared to a dedicated time-series model such as ARIMA.

- **6.3 Model training and forecasting**

The models were trained using historical COVID-19 case data. After training, forecasts were generated to predict future outbreak trends. The resulting predictions were analyzed through visualizations and performance metrics to assess model effectiveness.

ARIMA forecasts are based on patterns learned from past observations. Since the data showed a repeating weekly behavior, the model projected similar fluctuations into the forecast horizon rather than producing a straight line.

- **6.4 Model Serialization:**

After training, the ARIMA model was saved as a serialized .pkl file using Joblib. This allowed the trained model to be reused without retraining and enabled seamless deployment within the Streamlit dashboard.

CHAPTER 7: RESULTS AND EVALUATION

• 7.1 Stationarity Testing:

The Augmented Dickey-Fuller (ADF) test was performed to evaluate stationarity. The initial test on the original series produced a borderline p-value of 0.031, indicating the series was only marginally stationary and still carried trend characteristics typical of cumulative outbreak data. First-order differencing was applied to remove this trend. The ADF test was re-run on the differenced series, yielding a p-value of approximately 4.91×10^{-9} (ADF statistic = -6.66), confirming that the series became strongly stationary after differencing. This validated the use of $d = 1$ and confirmed the series was suitable for ACF/PACF analysis.

```

[22]  ## Stationarity Testing
      ##We use the Augmented Dickey Fuller (ADF) test to determine whether the time series is stationary before fitting an ARIMA model.
      from statsmodels.tsa.stattools import adfuller

[23]  result = adfuller(global_series)

[24]  print('ADF Statistic: %f' % result[0])
      print('p-value: %f' % result[1])

      ADF Statistic: -3.044629
      p-value: 0.030932

[26]  p_value = result[1]

[27]

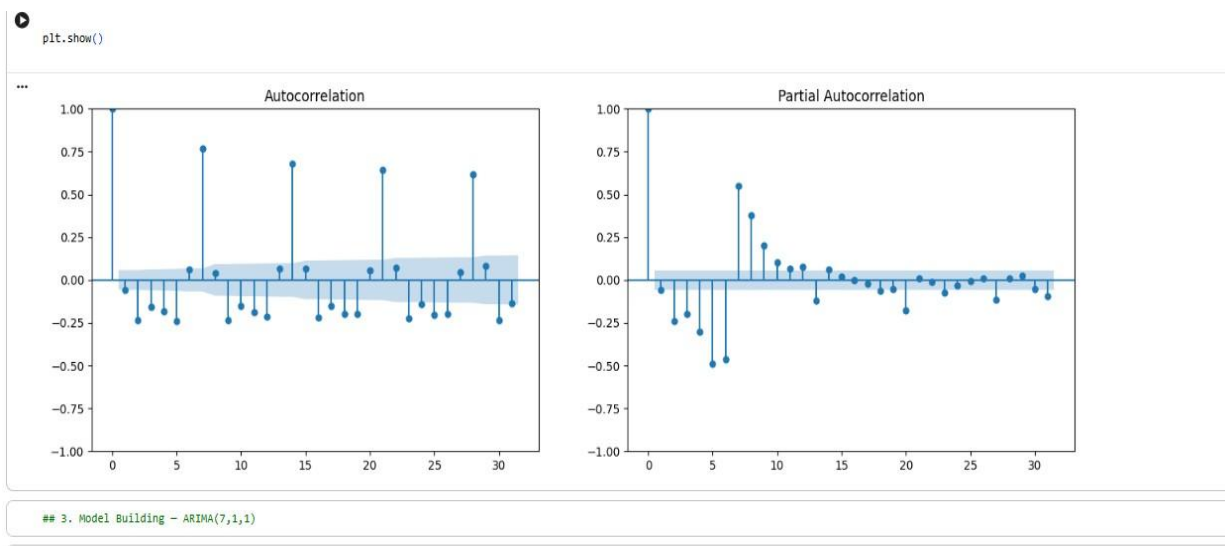
```


• 7.2 ACF and PACF Analysis:

The Autocorrelation Function (ACF) plot was used to identify significant correlations between observations at different lag values. Noticeable spikes indicated the presence of temporal dependencies and helped determine appropriate model parameters.

The Partial Autocorrelation Function (PACF) plot was analyzed to identify the direct influence of previous observations on current values. Significant lags observed in the plot assisted in selecting the autoregressive component of the ARIMA model.

After applying first-order differencing to the global confirmed cases series, the ACF and PACF plots were examined to determine the order of the ARIMA model. The Partial Autocorrelation Function (PACF) showed statistically significant spikes from lag 1 through lag 7, after which the values fell within the confidence interval, indicating an autoregressive order of $p = 7$. The Autocorrelation Function (ACF) did not show a significant spike at lag 1, suggesting that no moving average term was required, and $q = 0$ was selected.



- **7.3 ARIMA Performance:**

The ARIMA(7,1,0) model was evaluated using forecasting error metrics. The model achieved a Mean Absolute Error (MAE) of 34,609.48 and a Root Mean Squared Error (RMSE) of 44,536.42. Considering the large scale and variability of global daily outbreak cases (often in the hundreds of thousands), these error values indicate that the model was able to capture the general direction of the trend, though with notable deviation in magnitude. The difference between MAE and RMSE suggests the presence of occasional larger prediction errors, which is expected when working with real-world outbreak data characterized by sudden fluctuations, weekly reporting cycles, and reporting variations not fully captured by a non-seasonal model.

- **7.4 Linear Regression Performance:**

The Linear Regression model achieved an R^2 score of 0.608, indicating a moderate relationship between historical and future case counts. While the model captured general trends, it was less effective than ARIMA in handling time-series patterns and temporal dependencies.

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, shuffle=False)
```

```
ml_model = LinearRegression()
ml_model.fit(X_train, y_train)

y_pred = ml_model.predict(X_test)

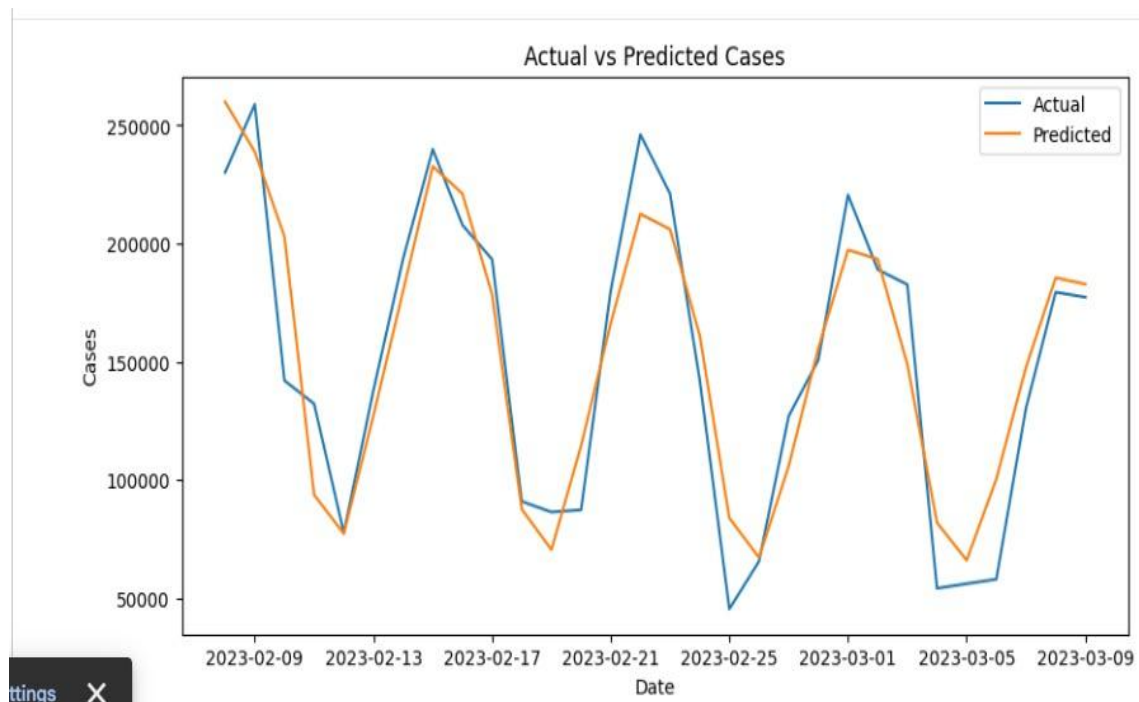
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print(f"Mean Squared Error: {mse}")
print(f"R-squared Score: {r2}")

Mean Squared Error: 30255999342.722622
R-squared Score: 0.6079622189255698
```

• 7.5 Actual vs Predicted Analysis:

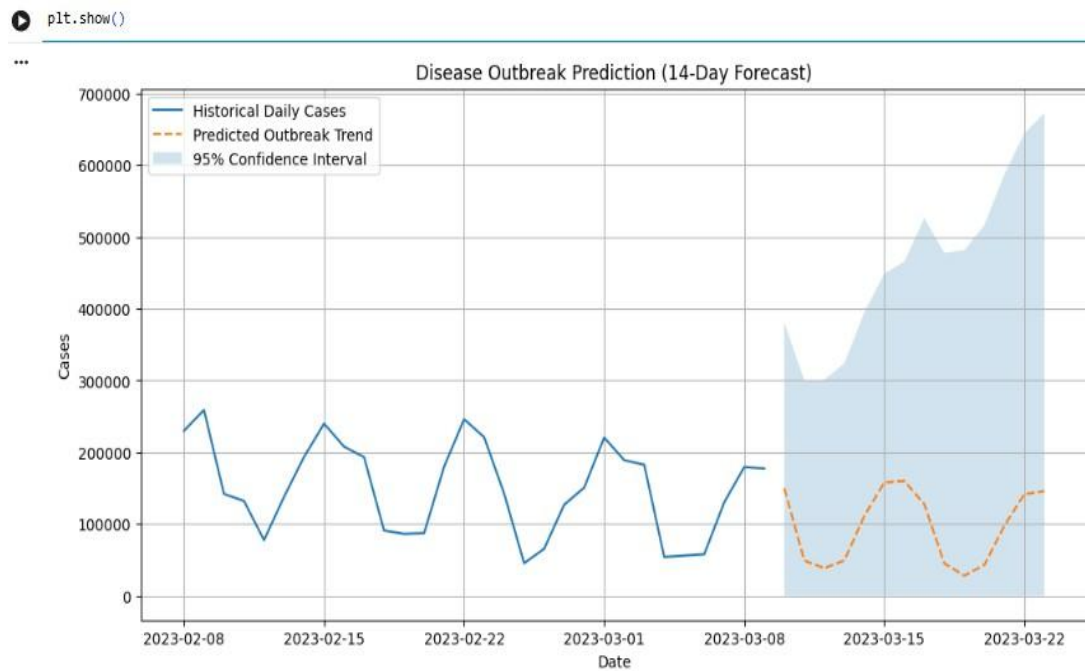
The Actual vs Predicted graph compares model predictions with observed case counts. The predicted values generally follow the trend of the actual data, indicating that the model successfully captured the underlying patterns within the dataset.



• 7.6 Forecast Results:

The ARIMA (7,1,0) model generated a 14-day forecast of future COVID-19 case trends. The forecast graph combines historical observations with future predictions and includes a 95% confidence interval to represent prediction uncertainty. The results demonstrate the model's ability to provide realistic short-term outbreak forecasts.

The forecast displays cyclical fluctuations because the underlying outbreak data contains recurring weekly patterns that were captured by the ARIMA model.

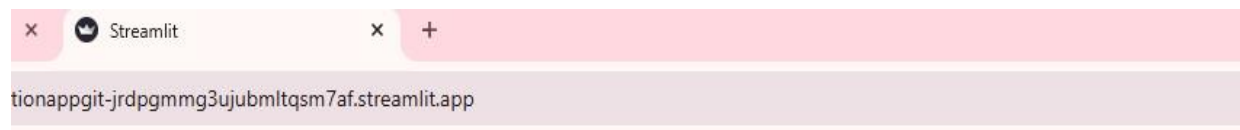


The historical data showed strong weekly seasonality. The ARIMA model captured this repeating pattern and projected it into future periods.

CHAPTER 8: DASHBOARD DEPLOYMENT

• 8.1 Streamlit Dashboard:

To improve accessibility, the forecasting model was deployed using Streamlit Cloud. The dashboard allows users to interactively select a forecasting horizon and visualize future outbreak predictions through charts and tables.



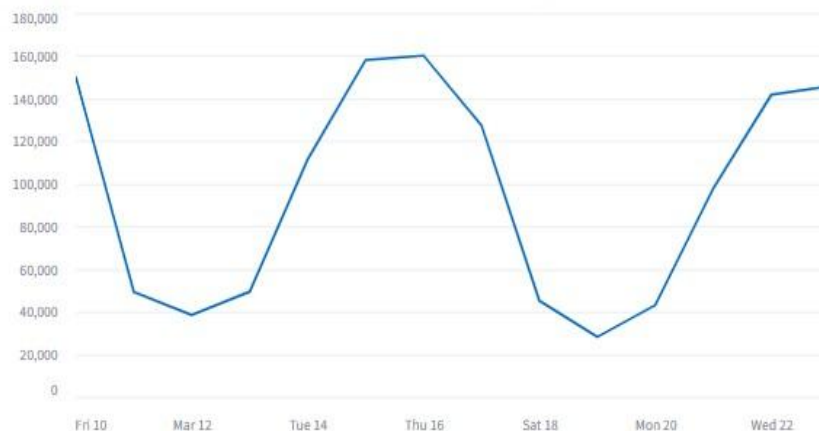
Disease Outbreak Prediction Dashboard

This tool uses an $ARIMA(7,1,1)$ model to forecast future global case counts.

Select the number of days to forecast:



Projected Outbreak for the Next 14 Days



- **8.2 Dashboard Features:**

The dashboard includes several useful features, including interactive forecast generation, dynamic visualizations, model integration through a serialized .pkl file, and an intuitive user interface that enables non-technical users to explore outbreak predictions.

- **8.3 Deployment Workflow:**

The deployment process involved training the ARIMA model, saving it as a .pkl file, uploading project files to GitHub, and connecting the repository to Streamlit Cloud. This workflow transformed the forecasting model into a publicly accessible web application.

CHAPTER 9: CHALLENGES FACED

Throughout the development of this project, several technical challenges were encountered. These challenges provided valuable learning opportunities and helped improve our understanding of data preprocessing, time-series forecasting, model evaluation, and deployment techniques.

1. Handling and preprocessing time-series data required careful date formatting and aggregation of daily case counts.
2. Selecting appropriate ARIMA parameters (p , d , q) was challenging and required analysis of ACF and PACF plots.
3. Ensuring stationarity of the time series was necessary before model training, which involved differencing the data.
4. Forecast visualization and interpretation required validation to ensure predictions aligned with historical trends.
5. Deploying the model using Streamlit involved managing dependencies and model file integration.
6. Evaluating model performance required comparing predicted values against actual observations using metrics such as MAE and RMSE.

CHAPTER 10: CONCLUSION

This study successfully analyzed global disease outbreak data and applied time-series forecasting techniques to predict future case trends. Through data preprocessing, exploratory data analysis, and model development, the ARIMA(7,1,0) model was used to generate forecasts based on historical patterns. The results demonstrated the effectiveness of data science techniques in understanding and forecasting outbreak behavior. Overall, this work provided valuable practical experience in data analysis, predictive modeling, and data visualization while highlighting the importance of data-driven decision-making.